

Microbiology: Skin and Mucous Membrane Infections

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- Session Objectives:
 1. Describe the roles of microbial virulence factors and host defense mechanisms play in the prevention of or development of skin and mucous membrane infections.
 2. Describe common medically important pathogens that cause skin and mucous membrane infection.
 3. Diagnose pathogens causing skin and mucous membrane infections.
- Connections
 - ScholarRx: [Bacterial Infections of the Skin](#)

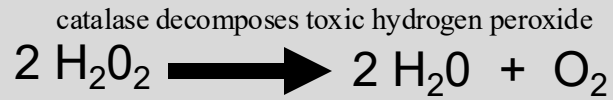


Normal Flora of the Skin

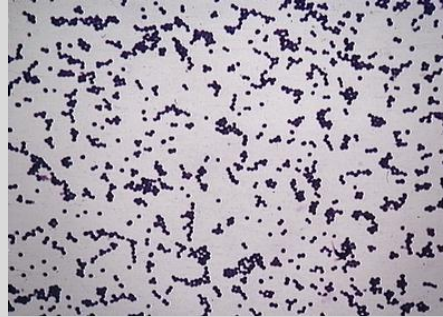
- Gram positives:
 - *Staphylococcus aureus* (30-60%), *S. epidermidis* (~100%)
 - *Streptococcus pyogenes* (Group A *Streptococcus* [GAS])
 - Diptheroids: *Corynebacterium* and *Propionibacterium* spp.
- Gram negatives: *Proteus* & *Pseudomonas* spp.
(transient)
- Fungi: *Candida albicans*, dermatophytes



Staphylococci

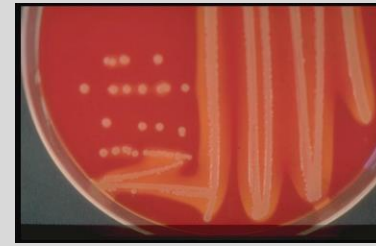


- Gram-positive catalase positive, non-motile cocci
 - 0.5 – 1.5 μm diameter
- Grows well on blood agar, high NaCl (10%) 18-40°C
 - Does not grow on MacConkey
- Facultative anaerobe
- May be sensitive or resistant to methicillin
 - e.g., MRSA = methicillin-resistant *Staphylococcus aureus*



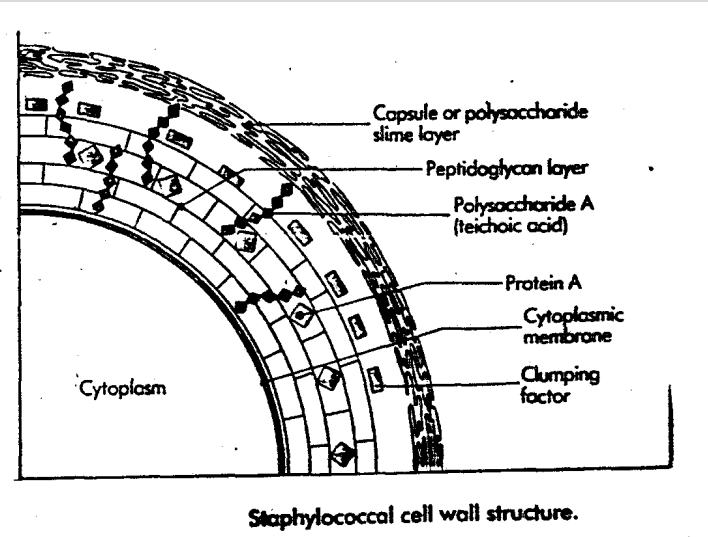
Staphylococcus aureus

- **Coagulase positive**, beta hemolytic on blood agar
- Key virulence factors
 - **Coagulase** converts fibrinogen to fibrin, clots and a fibrin shield inhibiting phagocytosis
 - **Capsule** inhibits opsonization and phagocytosis, protects from C'-mediated leukocyte destruction
 - **Protein A** binds IgG1, IgG2, IgG4 Fc receptors, inhibits opsonization, phagocytosis, leukocyte chemoattractant, anticomplement
 - **Catalase** prevents oxidative killing by immune cells
 - Hyaluronidase, fibrinolysin, lipase, nuclease
 - **Penicillinase** opens beta lactam ring inactivating penicillin
 - Cytolytic & exfoliative toxins, **TSST**, enterotoxin



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Staphylococcus aureus: Skin Diseases

- Furuncle (boil): infected hair follicle

- Carbuncle: coalesced cluster of furuncles



- Impetigo: skin infection via cut or scrape, toxin-mediated keratin separation, flaccid blisters



- Pyogenic wound infection

- Scalded skin syndrome

- Usually small children
- Spreading local inflammation, peeling of squamous epithelium, recovery
- **Nickolsky sign**: apply horizontal pressure to skin adjacent to lesion results in sheering off of top layers of epidermis



Staphylococcus aureus: Other Diseases

- Bloodstream infection
- Sepsis, septic shock
- Acute endocarditis
- Pneumonia
- Empyema
- Osteomyelitis
- Septic arthritis
- Toxic shock syndrome
 - Superantigen stimulates T-cell proliferation, cytokine production, endothelial destruction, capillary leak, shock, multiorgan failure
 - Strawberry tongue
 - Diffuse, erythematous rash



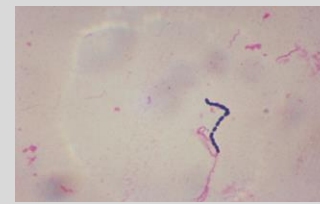
Coagulase-negative *Staphylococci*

- Catalase-positive, **coagulase-negative**, novobiocin-sensitive, no hemolysis on blood agar
- *Staphylococcus epidermidis*
 - Opportunistic pathogen
 - Slime layer facilitates adherence to foreign bodies (prosthetic joints, heart valves, intravenous catheters)
- *Staphylococcus saprophyticus*
 - UTI (cystitis) in sexually active women



Streptococci

- **Gram-positive**, non-motile cocci in pairs and chains, 0.5-1.2 μm
- Catalase-negative, facultative anaerobic, range from strict anaerobics to capnophilic (use CO_2 for growth)
- Mechanical factors
 - Polysaccharide capsule antiphagocytic
 - Hyaluronic acid in GAS
 - M protein (GAS) antiphagocytic, superantigen
 - F protein
 - Lipoteichoic acid
 - T protein
- Toxins
 - Pyrogenic exotoxins
- Erythrogenic toxin (scarlet fever rash)
- Enzymes
 - Streptolysin O (causes beta-hemolysis)
 - Streptolysin S
 - Streptokinase (A&B)
 - DNase (A-D)
 - C5a peptidase
 - Others



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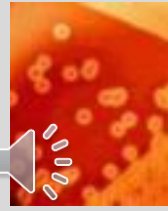
Biochemical Classification	Serologic Classification	Hemolysis Patterns
<i>S. pyogenes</i>	A	β
<i>S. agalactiae</i>	B	β ; occasionally nonhemolytic
<i>S. dysgalactiae</i>	C, G	β
<i>S. anginosus</i> group	A, C, F, G, nongroupable	β ; occasionally α or nonhemolytic
<i>S. bovis</i>	D	α ; nonhemolytic; occasionally β
Viridans group	Nongroupable	α or nonhemolytic
<i>S. pneumoniae</i>	Nongroupable	α



Gamma: no hemolysis



Alpha: Incomplete (causes green zones)



Beta: complete (causes clear zones)

Exudative pharyngitis (usually GAS)

- Sore throat, fever
- Diagnosis: rapid antigen, culture if negative, anti-streptolysin-O antibodies (ASO)
- Treatment: penicillin (1st line), amoxicillin, cephalexin, (anaphylaxis to PCN: clindamycin, clarithromycin)



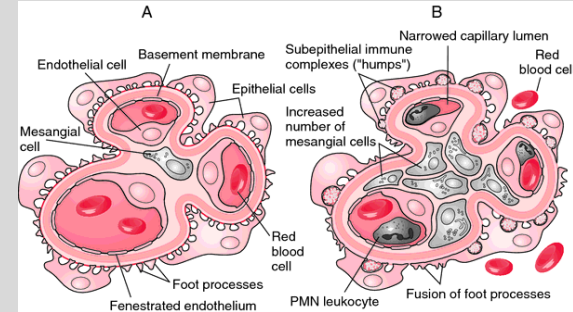
Streptococcal Skin Diseases

- Impetigo
 - Similar to staphylococcal impetigo, but bullae are tense and have gold crusts
- Scarlet fever
 - High fever, sore throat, sandpaper rash
 - Erythrogenic toxin gene encoded by lysogenic bacteriophage)
- Erysipelas
 - Acute, rapidly progressing, superficial skin infection
 - Bright red, warm, tender, sharply demarcated plaque with raised borders
- Cellulitis
 - Rapidly spreading, painful, erythematous skin and subcutaneous tissue infection
 - May result in bloodstream infection, endocarditis
- Necrotizing fasciitis
 - Rapidly advancing erythematous, warm bullae, crepitus (gas in tissues) and dusky skin discoloration
 - Intensely painful skin and soft tissue infection associated with fever, tachycardia, sepsis
 - May result in septic shock
 - Requires emergent surgical debridement
- Toxic shock syndrome
 - Similar to staphylococcal toxic shock



Streptococci: Non-skin MM diseases

- Pneumonia (*S. pneumoniae*)
- Meningitis
- Post-streptococcal sequelae (GAS)
 - Rheumatic fever (GAS pharyngitis or scarlet fever)
 - Post-streptococcal acute glomerulonephritis (GAS pharyngitis)



Other Clinically Important *Streptococci*

- *Streptococcus agalactiae*

- Neonatal bloodstream infection and meningitis
- Post-partum sepsis (colonizes birth canal)

- *Streptococcus pneumoniae*

- Pneumonia, sinusitis, otitis media, meningitis, bloodstream infection, sepsis

- Viridans streptococci

- Endocarditis, dental infections



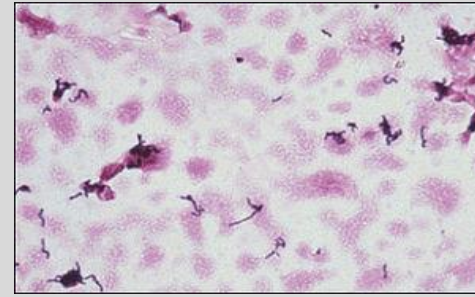
Pseudomonas aeruginosa

- Gram-negative, motile, oxidase-positive, non-lactose fermenting bacillus
- Colorless colonies on MacConkey agar
- Ubiquitous in moist environments
- Cell wall lipopolysaccharide (LPS) acts as an endotoxin resulting in sepsis
- Exotoxin A and esoenzyme S inhibit protein synthesis
- Infections of burns, surgical wounds, cystic fibrosis, ecthyma gangrenosum in neutropenia



Propionibacterium acnes

- Gram-positive, small, anaerobic (aerotolerant)
- Colonizes skin, conjunctiva, external ear, oropharynx
- Major cause of acne in young adults
 - Combines with sebum to stimulate host inflammatory response causing acne tissue damage



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Actinomyces israelii

- Gram-positive, elongated rods develop filamentous forms
- Colonize URT, granulomatous sinusitis, lumpy jaw, pelvic/intrauterine abscess
- Infection associated with sulfur granules

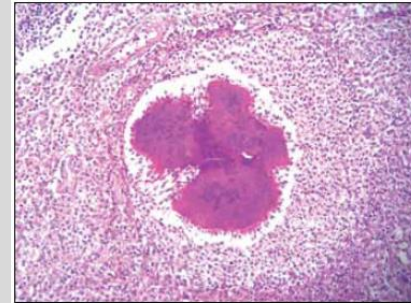
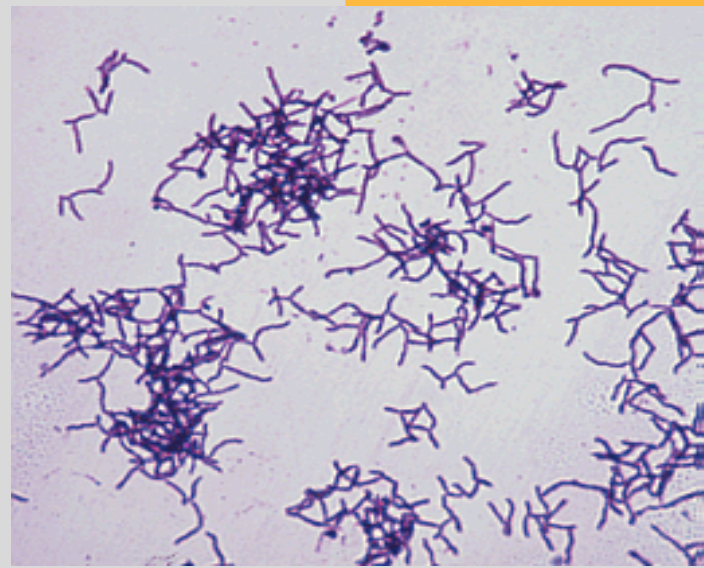


Figure 3: Characteristic sulfur granules of *Actinomyces* in the microabscess (H&E, x10).



Mycobacterium ulcerans: Buruli ulcer

- Acid fast bacillus related to *M. tuberculosis*
- Tropical Africa and Asia
- Associated with aquatic environments
- Large, erythematous ulcer



Parvovirus B19 (Fifth Disease)

- Non-enveloped, icosahedral, single-stranded, linear DNA virus
- Causes erythema infectiosum
 - Childhood febrile illness
 - Slapped-cheek rash



Thank you for reviewing and good luck!

- Summary

1. The normal flora of the skin is predominantly Gram-positive *staphylococci* and *streptococci*
2. Skin flora and pathogens can cause various infections of the skin as well as toxin-mediated illness
3. Biochemical tests can help identify pathogens to direct therapy

- [Feedback](#)

